

Physics Factsheet

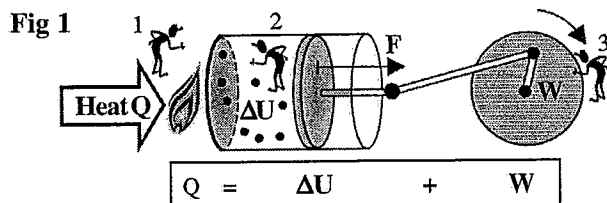


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Number 128

Calculations: First Law of Thermodynamics

The **first law of thermodynamics** can be explained by Fig 1 which shows a frictionless cylinder and piston containing a quantity of gas.



Three imps are watching what is happening. The first is aware of a quantity of heat (Q) entering the cylinder, the second imp doesn't understand 'heat' but sees the gas molecules speed up while the third sees work output (W) at the turning wheel. All three imps understand the term *energy*. They decide that the heat *energy* input must equal the *energy* gained by the gas molecules plus the *energy* used when the work is done.

So, if 100J of heat enter the gas and 70J of energy are given to the gas molecules then the remaining 30J must appear as the work done.

Molecular kinetic energy and Temperature

The second imp is keeping an eye on the gas molecules. He sees the molecules speed up and *gain kinetic energy*. When you or I look at the gas we see that there is a *gain in temperature*.

Key: When the temperature increases we deduce that the molecular kinetic energy has also increased.

We use U to represent the kinetic energy of all the molecules and T for temperature and write $U \propto T$
Remember that the temperature must be in kelvin K .

Example 1

A gas has a temperature of 27°C . Find the temperature when the kinetic energy of the gas is doubled.

Answer. The temperature $T = 27 + 273 = 300\text{K}$. To double the kinetic energy we must double the kelvin temperature. So the required temperature is 600K or $(600 - 273) = 327^\circ\text{C}$.

Another name for the kinetic energy of the gas molecules is the *internal energy*.... this term is used in thermodynamics. In the example above, the internal energy is doubled and in question 1 (at the end), the gas loses 10% of its internal energy.

Remember that the second imp 'sees' a change in internal energy while you and I 'see' a change in temperature.

Now look again at Fig 1.

The heat that flows into the gas has two jobs to do.

- 1) it must warm the gas ie increase the internal energy and
- 2) it must provide for work to be done

Key: We write $Q = \Delta U + W$. This is the first law of thermodynamics

Q is the heat supplied to the system, ΔU is the rise in internal energy of the system, and W is the work done by the system.

In our first example, the heat $Q = 100\text{J}$, the increase in internal energy $\Delta U = 70\text{J}$ and the work done $W = 30\text{J}$

This illustrates an important point about heat and work. If we rub our hands together, work is done and heat produced. In this example, *all* of the work done against friction is converted into heat.

Work $\rightarrow$ Heat (100% conversion)

If we try the *reverse* process, Heat $\rightarrow$ Work, we find that 100% conversion is impossible. This is not due to quality of workmanship in making the cylinder and piston but is down to the laws of physics! Some of the heat supplied is used to warm the gas (ΔU). Only that left over is available for work to be done.

Example 2

For the example in fig 1 ($Q = 100\text{J}$, $\Delta U = 70\text{J}$ and $W = 30\text{J}$), calculate the efficiency of the cylinder and wheel arrangement. In this example we are trying to convert the input (100J of heat) into the output (30J of work).

$$\text{efficiency} = \frac{\text{output}}{\text{input}} = \frac{30\text{J}}{100\text{J}} = 30\%$$

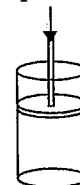
Adiabatic Changes

You may have noticed when pumping up a tyre, that the pump gets quite warm. This is not just due to friction but because you are *compressing* the gas. You are doing work on the gas.

Imagine that the cylinder in Fig 2 is perfectly insulated so that heat cannot enter nor leave the gas.

When the piston is rapidly pushed inwards, there is a temperature rise in the gas. How can this be? The temperature increases but heat is not supplied.

Fig 2. rapid compression



The answer is found from the first law, $Q = \Delta U + W$. Suppose that 200J of work are done in *compressing* the gas.

Key: Because the gas is *compressed* we must take the work done to be *negative*. Heat doesn't enter or leave the gas so $Q = 0$. Substitute in the equation:- $Q = \Delta U + W$
 $0 = \Delta U + (-200\text{J})$
 $200\text{J} = \Delta U$

The internal energy has increased by 200J

Remembering that internal energy depends on temperature, we see that the temperature must also increase.

This is an example of an **ADIABATIC CHANGE**, ie, where heat is *not* allowed into or out of the gas.

For a compression, there is **no** heat exchange but there is a temperature rise! Put another way, all the work done goes to increase the internal energy (and the temperature) of the gas.

Isothermal Changes

Look at Fig 3 showing a gas expanding at a *constant temperature*. This is an **Isothermal Change**.

Because the temperature doesn't change, the internal energy doesn't change. ie $\Delta U = 0$

This time 400J of work are done. Because the gas is *expanded* we must take the work done to be *positive*.

Substitute in the equation

$$\begin{aligned} Q &= \Delta U + W \\ Q &= 0 + 400\text{J} \\ Q &= 400\text{J} \end{aligned}$$

For the change to be isothermal, it must take place slowly to allow heat into or out of the gas. We have just shown that because 400J of work are done isothermally, the heat supplied must *also* be 400J.

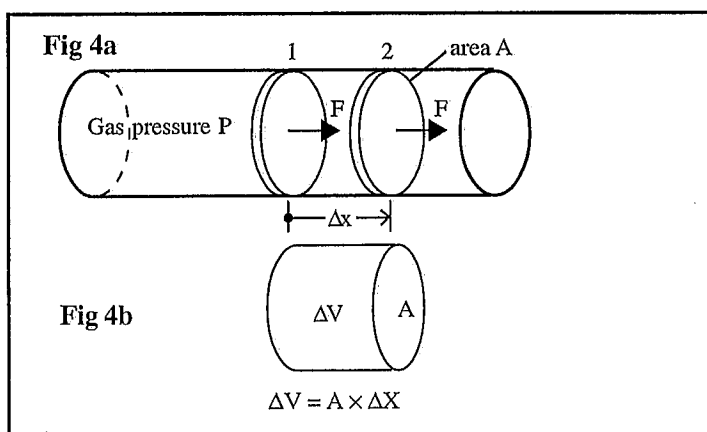
Summary

The First Law of Thermodynamics says $Q = \Delta U + W$ where Q is the heat absorbed, ΔU is the increase in internal energy and W is the work done by the system.

When the gas expands W is positive; for a compression W is negative. If the heat enters the gas, Q is positive. If the heat leaves the gas, Q is negative. $U \propto T$ (kelvin temperature). If U increases, then the temperature increases (and vice versa).

Work done by a gas

Fig 4a shows some gas trapped in a cylinder by a piston.



As the gas expands, the force F pushes the piston through a small distance, so we conclude that **work** is done! We have to find the work done by the gas as the piston moves a small distance Δx from position 1 to position 2.

The gas is at a pressure P and exerts a force F on the piston with area A .

$$\text{Pressure} = \frac{\text{force}}{\text{area}} \quad P = \frac{F}{A} \Rightarrow F = P \times A$$

The distance Δx is small so the pressure remains constant. The work done by the force F is also small and is ΔW .

Work done = Force $\times$ Distance, so

$$\Delta W = F \times \Delta X$$

$$\Delta W = (P \times A) \times \Delta X = P \times (A \times \Delta X)$$

From Fig 4b we can see that the *increase* in volume $\Delta V = A \times \Delta X$. Hence we have the important result

$$\Delta W = P \times \Delta V$$

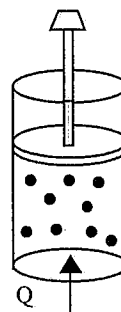
You should check the units of this equation. The units of P are Nm^{-2} ; those for ΔV are m^3 .

Using the equation above you should show that the unit for ΔW is the Joule.

Example 3

Fig 5 shows a gas being heated at a constant pressure of 10^5 Pa (atmospheric). The volume of the gas increases by 0.5 l . Find the work done.

Fig 5



When we use $\Delta W = P \times \Delta V$, we want W in **JOULES**.

This means that P must be in Pa (Nm^{-2})

and V must be in m^3 .

Because $10^3 \text{ litres} = 1 \text{ m}^3$, $0.5 \text{ l} = 0.5 \times 10^{-3} \text{ m}^3$ so we write $\Delta V = 0.5 \times 10^{-3} \text{ m}^3$.

$$\Delta W = P \times \Delta V$$

$$\Delta W = 10^5 \times (0.5 \times 10^{-3})$$

$$\Delta W = 50 \text{ J}$$

The work done by the expanding gas is 50 J.

Remember that pressure is sometimes given in kPa or MPa , so you must then introduce 10^3 or 10^6 respectively in your calculation.

Exam Hint: Remember to convert the volume into m^3 and the pressure into Pa (Nm^{-2}) before performing the calculation.

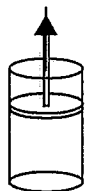
If in this case, (Fig 5), the heat supplied was 125J, use the first law to find the gain in internal energy (ΔU) of the gas.

We have $Q = \Delta U + W$, with $Q = 125\text{J}$, and $W = 50 \text{ J}$.

The gain in internal energy $\Delta U = (125\text{J} - 50\text{J}) = 75\text{J}$

Practice Questions

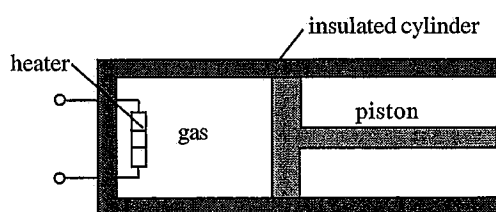
1. A quantity of gas has a temperature of 27°C and loses 10% of its kinetic energy. Calculate its initial and final kelvin temperature.
2. The diagram below shows an adiabatic expansion.....no heat gain or loss.



This time 300J of work are done. Because the gas is *expanded* we must take the work done to be *positive*.

Calculate:

- (a) the change in internal energy.
 - (b) has the temperature gone up or down?
3. A gas is heated at a constant pressure of 10^6 Pa and its volume increases by 8 litres. Calculate the work done by the gas as it expands. If the heat energy supplied to the gas is 10kJ, calculate the gain in internal energy of the gas.
 4. An engine discharges 15m^3 of hot gas at a pressure of 1.0×10^5 Pa to the atmosphere that is also at a pressure of 1.0×10^5 Pa. The gas then cools and contracts to one third of its original volume.
 - (a) Calculate the work done on the exhaust gas.
 - (b) During the cooling process, 5.0 MJ of heat is transferred to the atmosphere. Calculate the change in internal energy of the gas.
 5. The diagram shows an insulated cylinder fitted with a perfectly fitting frictionless piston. The cylinder contains a fixed mass of gas and a heater.



Two experiments are performed:

Experiment 1

The heater provides 150 J of energy with the piston held in a fixed position. The temperature rise of the gas is 30K.

Experiment 2

The heater again supplies 150J of energy with the piston free to move so that the gas expands at constant pressure and does some useful work W . In this case, the temperature rise is 18K and the efficiency is 40%

- (a) Explain why there is a difference in the temperature rise in each case.

In the first experiment, calculate

- (b) the gain in internal energy of the gas

In the second experiment, calculate

- (c) the work done by the expanding gas

- (d) the gain in internal energy of the gas

In the second case, the gas expanded at a constant pressure of 1.0×10^5 Pa, calculate

- (e) the change in volume of the gas.

Answers

1. The kinetic energy (internal energy) $\propto$ kelvin temperature.
If 10% of kinetic energy lost then 10% of kelvin temperature 'lost'.
 $27^{\circ}\text{C} = 300\text{K}$. 10% loss gives 270K $(270-273)^{\circ}\text{C} = -3^{\circ}\text{C}$
2. In all cases, $Q = \Delta U + W$.
For adiabatic change, $Q = 0$. Gas expands so work done is Positive.
 $0 = \Delta U + 300\text{J}$
 $\Delta U = -300\text{J}$. That is a decrease in internal energy with a corresponding decrease in temperature.
3. $P = 10^6$ Pa, $\Delta V = 8\text{ l} = 8 \times 10^{-3} \text{ m}^3$.
Work Done $= P \times \Delta V$
 $= 10^6 \times 8 \times 10^{-3} = 8 \times 10^3 \text{ J} (8\text{kJ})$
 $Q = \Delta U + W$
 $10\text{kJ} = \Delta U + 8\text{kJ} \Rightarrow \Delta U = 2\text{kJ}$ (the increase in internal energy)
4. (a) Work done $= P \times \Delta V$
 $= 1.0 \times 10^5 (15-5)$
 $= 1.0 \times 10^6 \text{ J}$
This is a compression so $W = -1.0 \times 10^6 \text{ J}$
(b) The exhaust gas is the system, it loses $5.0 \times 10^6 \text{ J}$
and so, $Q = -5.0 \times 10^6 \text{ J}$
First Law, $Q = \Delta U + W$
 $-5.0 \times 10^6 = \Delta U + (-1.0 \times 10^6)$
 $-4 \times 10^6 = \Delta U$
i.e., a loss in internal energy of 4MJ
5. (a) In the first part the piston is fixed so no work is done. All the 150J increases the internal energy and consequently the temperature. In the second part the piston is free to move, work is done, so now less energy is available to increase the internal energy with a corresponding lower temperature rise.
(b) $Q = \Delta U + W$, because $W = 0$, $Q = \Delta U = 150\text{J}$
All the energy supplied (150J) increases the internal energy
(c) efficiency = work / energy supplied
 $40/100 = W/150$. $\Rightarrow W = 60\text{J}$
(d) $Q = \Delta U + W \Rightarrow 150\text{J} = \Delta U + 60\text{J} \Rightarrow \Delta U = 90\text{J}$
(e) $W = P \times \Delta V$
 $60 = 1.0 \times 10^5 \times \Delta V$
 $\Delta V = 60 \times 10^{-5} \text{ m}^3$ or 0.6 litres

Acknowledgements:

This Physics Factsheet was researched and written by Keith Cooper
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ISSN 1351-5136

